

A Regime-Aware Model for Loan-Level Probability of Default

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Abstract

This study develops and evaluates an interpretable regime-aware model for loan-level mortgage probability of default (PD). Monthly Freddie Mac origination and servicing records are combined with public macroeconomic indicators from the Federal Reserve Economic Data database. A two-state Gaussian hidden Markov model compresses housing, refinancing, labour-income, and financial-conditions information into a filtered macroeconomic stress probability. That probability enters a regularised logistic PD model both directly and through interactions with borrower and payment-history characteristics. The empirical design uses a twelve-month transition to serious delinquency, chronological training, validation, and test samples, and validation-based probability calibration. On 4,106,275 held-out loan-month observations, the primary Platt-calibrated regime-aware model records a Brier score of 0.006015, log loss of 0.032945, ROC-AUC of 0.827256, and PR-AUC of 0.160097. The corresponding borrower-only model records 0.006055, 0.033106, 0.821979, and 0.148361. These test-period gains are modest and do not extend uniformly to validation or monthly portfolio-count calibration. Moreover, the inferred stress probability is nearly saturated during the test period. The present draft therefore treats the regime-aware result as incremental and conditional evidence, and identifies rolling-origin estimation and regime-signal robustness as necessary publication checks.

Keywords: probability of default; mortgage credit risk; hidden Markov model; regime switching; calibration; logistic regression; macroeconomic stress

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Introduction

1.1 Probability of Default as a Loan-Level Quantity

Probability of default (PD) is the conditional probability that an obligor enters a contractually or regulatorily defined default state during a specified future horizon. In retail mortgage portfolios, PD is not merely a ranking score. It is a probability used to support underwriting, account monitoring, pricing, expected-loss measurement, stress testing, and capital planning. A useful PD model must therefore answer two questions simultaneously: which loans are riskier than others, and whether a group assigned a probability such as one percent actually defaults at approximately that rate. The first question concerns discrimination; the second concerns calibration (Brier, 1950; Gneiting and Raftery, 2007; Van Calster et al., 2019).

For loan i observed in month t , this study defines the forward twelve-month serious-delinquency outcome

$$Y_{i,t}^{(12)} = \mathbb{I}\{\text{loan } i \text{ first reaches at least 90 days past due in } (t, t + 12]\}, \quad (1.1)$$

where $\mathbb{I}\{\cdot\}$ is the indicator function. The target PD is

$$\pi_{i,t}^{(12)} = \Pr\left(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t\right), \quad (1.2)$$

and $\mathcal{F}_t$ denotes information available at month t . Equations (1.1) and (1.2) make three de-

sign commitments explicit. The prediction unit is a loan-month; the event horizon is fixed at twelve months; and the model must avoid using information that becomes available only after the prediction date. The 90-days-past-due threshold follows the serious-delinquency convention used in mortgage performance analysis and is consistent with the broad Basel notion that material delinquency is evidence of default (Basel Committee on Banking Supervision, 2004; Board of Governors of the Federal Reserve System, 2006).

1.2 Why Static PD Relationships May Fail

Traditional credit scorecards estimate default risk from borrower and contract characteristics such as credit score, debt-to-income ratio, loan-to-value ratio, interest rate, balance, occupancy, and recent payment performance. Logistic regression remains a central benchmark because its coefficients have an odds-ratio interpretation, its probability mapping is explicit, and its behaviour can be audited more readily than that of many high-capacity learners (Hand and Henley, 1997; Thomas, 2000; Costa e Silva et al., 2020). These properties matter in credit decisions, where predictive power must be balanced against stability, governance, and the ability to explain why risk changes.

The static scorecard nevertheless imposes an economically strong restriction: the relationship between a borrower characteristic and default is constant through calendar time. That restriction is questionable for mortgages. High leverage is more consequential when collateral values weaken; a high payment burden is more dangerous when employment conditions deteriorate; and an adjustable or elevated mortgage rate is more difficult to absorb when refinancing becomes expensive. Mortgage default is consequently shaped by both borrower-specific vulnerability and common economic conditions (Deng et al., 2000; Elul et al., 2010; Campbell and Cocco, 2015; Gerardi et al., 2018).

Observed macroeconomic covariates provide one solution. Unemployment, inflation, interest rates, house-price growth, the yield curve, and financial conditions can be added

directly to a loan-level PD equation. Survival and stress-testing studies show that such covariates can connect borrower models to the credit cycle (Bellotti and Crook, 2009, 2013; Figlewski et al., 2012). Yet an additive macro model assumes that the same observed variables have stable effects and that stress can be represented adequately by a linear combination of contemporaneous indicators. Economic stress is instead multidimensional and persistent. Inflation and policy tightening may dominate one episode, while labour-market deterioration, declining collateral values, or financial tightening may dominate another.

1.3 Research Gap

Three literature streams approach this problem but do not fully resolve it at the loan level. First, conventional consumer-credit and mortgage PD models provide borrower-specific probabilities but often impose time-invariant coefficients. Second, macro-augmented hazard or logistic models incorporate the economy but frequently treat macro effects as additive rather than allowing borrower sensitivities to change with the state of the economy. Third, regime-switching credit-risk models represent persistent latent conditions, but many applications focus on corporate default intensities, rating migrations, traded credit instruments, or portfolio aggregates rather than an interpretable mortgage loan-month score (Bangia et al., 2002; Yu et al., 2018; Milidonis and Chisholm, 2024).

Recent machine-learning work can discover nonlinear interactions and achieve strong ranking performance, including in mortgage portfolios (Khandani et al., 2010; Fuster et al., 2022; Kundig and Sigrist, 2025). However, flexibility alone does not answer the methodological question posed here. A model may rank defaults accurately while producing unstable or biased PD levels, and post-hoc explanations do not always substitute for a model whose structure is interpretable by construction (Rudin, 2019; Bussmann et al., 2021). The gap is therefore not the absence of predictive algorithms. It is the absence of a focused, calibration-first test of whether a probabilistic macro-regime signal adds reliable information

to a transparent loan-level mortgage PD model.

1.4 Aim, Research Questions, and Contribution

The aim of the study is to develop and empirically evaluate an interpretable regime-aware model for loan-level mortgage PD. A two-state Gaussian hidden Markov model (HMM) is used to infer a monthly probability of macroeconomic stress from public macroeconomic indicators. The filtered stress probability then enters a regularised logistic model through a level effect and selected interactions with borrower, loan, and payment-history variables.

The study addresses four research questions:

1. Does an HMM-derived stress probability improve held-out PD calibration relative to a borrower-only logistic model?
2. Does the latent stress signal add information beyond observed macroeconomic covariates?
3. Do selected borrower and payment-history sensitivities vary systematically with macroeconomic stress?
4. Are any gains stable across validation periods, test periods, calibration methods, and portfolio-level default-count diagnostics?

The contribution is deliberately narrower than a claim that regime switching is new in credit risk. The methodological contribution is a transparent bridge between a latent macroeconomic state model and a loan-level logistic PD model. The empirical contribution is a large-sample comparison using Freddie Mac mortgage performance records and public Federal Reserve Economic Data (FRED). The evaluative contribution is to judge the regime layer primarily by proper probability scores and calibration evidence, with ROC-AUC and PR-AUC treated as complementary measures rather than decisive proof of PD quality.

1.5 Study Design and Scope

The empirical workflow contains five linked stages. Freddie Mac origination and monthly servicing files are first normalised into a loan-month panel. Second, serious-delinquency labels and interpretable payment-history features are constructed. Third, public macroeconomic indicators are transformed into housing/collateral, rate/refinancing, labour/income, and financial-conditions factors, which enter a two-state Gaussian HMM. Fourth, borrower-only, macro-augmented, stress-only, and interaction-based logistic specifications are estimated on chronological samples with validation-based regularisation and probability calibration. Fifth, performance is evaluated using the Brier score, log loss, ROC-AUC, PR-AUC, calibration bins, Brier decomposition, time slices, stress-probability slices, and monthly predicted-default-count error.

The study intentionally retains an interpretable model class. Black-box algorithms are reviewed as relevant alternatives, but they are not the primary empirical benchmark. This boundary permits a direct assessment of the economic and statistical value of the regime mechanism rather than allowing an increase in generic model capacity to obscure the comparison.

1.6 Organisation of the Integrated Manuscript

Chapter 2 develops an extensive, self-contained review of loan-level PD modelling, mortgage default mechanisms, macroeconomic credit-risk models, latent regimes, calibration, and interpretability. It is written so that it can form the core of a review-paper submission after journal-specific extraction. Chapter 3 specifies the data construction, HMM, regime-aware logistic model, estimation, calibration, and evaluation procedures. Chapter 4 reports the current empirical evidence and its limitations. A final conclusion is intentionally omitted from this draft pending supervisor review and completion of the rolling-origin robustness analysis.

Literature Review

Review-Paper Front Matter

Working title: *From Static Credit Scores to Regime-Aware Probability of Default: A Critical Review of Loan-Level, Macroeconomic, and Latent-State Models*

Review abstract. Credit defaults are realised at borrower or loan level but vary with the macro-financial environment. This creates a modelling tension between granular credit scorecards, which are often interpretable but temporally static, and flexible state-dependent methods, which can capture economic change but may weaken calibration or transparency. This critical narrative review synthesises the mathematical and empirical literature on probability-of-default definitions, structural and reduced-form credit models, statistical credit scoring, mortgage default mechanisms, macroeconomic covariates, hidden Markov regimes, machine learning, explainability, and probability calibration. Particular attention is given to the information-set distinction between filtered and smoothed regime probabilities and to the difference between discrimination and calibrated PD estimation. The evidence indicates that latent regimes are best treated as a parsimonious common-state layer rather than a replacement for borrower and payment-history information. It also shows that credible regime-aware evidence requires chronological and rolling-origin validation, economically interpretable interaction effects, proper scoring rules, calibration sensitivity, and portfolio-level timing diagnostics. The review identifies a focused research gap: a transparent loan-month model in which an HMM-derived stress probability changes selected borrower sensitivities

and is evaluated primarily as a probability forecast. This framework offers a tractable bridge between static scorecards and black-box macro-sensitive prediction.

Review keywords: credit scoring; probability calibration; mortgage default; macroeconomic regimes; hidden Markov model; interpretability; stress testing

2.1 Scope and Organising Question

The literature on credit-risk modelling is unified by a scale problem. Default is realised by a particular borrower or loan, but its probability evolves with portfolio composition, lending standards, collateral markets, and the macro-financial environment. A useful loan-level probability-of-default (PD) model must therefore preserve borrower heterogeneity while responding to common economic conditions. The organising question for this review is: *how can macroeconomic state dependence be introduced into loan-level PD estimation without sacrificing probability calibration or structural interpretability?*

This chapter is a critical narrative review rather than a formal systematic review. It combines foundational work with recent studies that define the current methodological frontier. The foundational literature is necessary because structural models, reduced-form intensities, logistic scorecards, hidden Markov models (HMMs), and proper scoring rules supply the mathematical components of the proposed framework. Recent work is emphasised when assessing machine learning, explainability, mortgage portfolio modelling, and regime-aware credit forecasting, consistent with the expectation that a review article should engage closely with developments from the preceding five years (Bussmann et al., 2021; Shi et al., 2022; Noriega et al., 2023; Milidonis and Chisholm, 2024; Kundig and Sigrist, 2025; Brookfield et al., 2026).

The review proceeds from the definition of PD to model classes and then to evaluation. Sections 2.2–2.4 examine the probability target and the economics of mortgage default. Sections 2.5–2.7 examine observed macroeconomic covariates and latent regimes. Sections 2.8

and 2.9 address predictive flexibility, interpretability, and calibration. The final sections synthesise the evidence into a set of design requirements for an interpretable regime-aware loan-level model.

2.2 Conceptual and Mathematical Foundations of PD

2.2.1 Default as a Conditional Probability

PD is horizon-specific and information-set-specific. Let $Y_{i,t}^{(h)}$ denote whether loan i defaults during the next h months, conditional on being non-defaulted at month t . The object of interest is

$$\pi_{i,t}^{(h)} = \Pr\left(Y_{i,t}^{(h)} = 1 \mid X_{i,t}, \mathcal{M}_t\right), \quad (2.1)$$

where $X_{i,t}$ is the borrower and loan information set and $\mathcal{M}_t$ is macroeconomic information available at t . Equation (2.1) distinguishes PD from an unconditional historical default rate. A historical rate averages across borrower compositions and macroeconomic states; a conditional PD maps current information to a future event probability.

The conditioning set matters operationally. An application score may use information available at origination, whereas a behavioural score incorporates payment history observed after origination. Mortgage monitoring naturally creates a panel in which a loan contributes repeated monthly observations. The model must account for changing balances, seasoning, delinquency history, refinancing incentives, and calendar conditions. The same loan can therefore have different PDs at different months without any contradiction.

The event definition also matters. Accounting, regulation, and servicing systems can use different thresholds. Serious mortgage delinquency is commonly represented by 90 or more days past due, while broader default definitions may also include unlikely-to-pay events, foreclosure, bankruptcy, or charge-off. A publication-quality analysis must state the threshold, prediction horizon, risk-set condition, and censoring rule because changes in any of these al-

ter the estimand. This thesis uses the twelve-month transition to 90+ days past due among observations not already seriously delinquent.

2.2.2 Structural and Reduced-Form Traditions

Structural credit-risk models explain default through the economic value of the obligor’s assets. In the Merton framework, corporate equity is an option on firm assets and default occurs when asset value is insufficient to meet promised debt (Merton, 1974). This theory gives a direct economic role to leverage, volatility, and maturity. Its conceptual relevance to mortgages is the option-like role of collateral and borrower equity: negative equity can make default economically attractive, although household liquidity constraints, moving costs, recourse rules, and non-financial considerations prevent a mechanical mapping from property value to default.

Reduced-form models treat default as a stochastic event whose intensity depends on observable or latent information (Jarrow and Turnbull, 1995; Lando, 1998; Duffie and Singleton, 2003). If $\lambda_i(u)$ is a conditional default intensity, the survival probability over $[t, t+h]$ is often represented as

$$S_i(t, t+h) = \Pr(T_i > t+h \mid T_i > t, \mathcal{F}_t) = \exp\left[-\int_t^{t+h} \lambda_i(u) du\right], \quad (2.2)$$

under the usual intensity construction. The corresponding horizon PD is $1 - S_i(t, t+h)$. Dynamic hazard models extend this logic by allowing covariates to vary with time. Shumway (2001) demonstrates that treating observations dynamically can improve failure prediction relative to static classification, and Duffie et al. (2007) incorporates stochastic firm-specific and macroeconomic covariates into multi-period default prediction.

The distinction between structural and reduced-form models is useful but not absolute in empirical retail credit. Loan-level logistic and hazard models typically use economically meaningful covariates without claiming that those covariates form a complete structural

model. The present thesis follows that pragmatic tradition. It uses a probabilistic state model for the economy and an interpretable conditional probability model for the loan, rather than attempting to price a contingent claim or recover a full household utility model.

2.3 Statistical Credit Scoring and Loan-Level PD

2.3.1 From Classification to Probability Estimation

Early empirical credit models emphasised the classification of distress using financial ratios and discriminant analysis (Altman, 1968). Consumer-credit research subsequently developed a broad family of scorecards, including logistic regression, discriminant methods, decision trees, neural networks, and ensemble methods. Reviews by Hand and Henley (1997), Thomas (2000), and Crook et al. (2007) show that credit scoring is not a single algorithm but a decision system connecting data, model estimation, acceptance policy, performance monitoring, and economic loss.

The binary logistic model remains a standard benchmark:

$$\text{logit}(\pi_{i,t}) = \log\left(\frac{\pi_{i,t}}{1 - \pi_{i,t}}\right) = \beta_0 + X_{i,t}^\top \beta. \quad (2.3)$$

Equation (2.3) is attractive because $\exp(\beta_j)$ is the multiplicative change in the odds associated with a one-unit increase in feature j , holding other features constant. Standardisation and penalisation can stabilise estimates when features differ in scale or are correlated. The logistic link also produces an explicit probability, unlike a raw classification margin.

The model has limitations. Linearity on the log-odds scale may miss nonlinear effects, interactions must be specified, and rare events can complicate estimation. Even so, logistic regression often remains competitive when data preparation, feature design, validation, and calibration are handled carefully (Costa e Silva et al., 2020). Benchmark studies repeatedly show that algorithm rankings depend on the dataset and objective rather than yielding a

universally superior classifier (Lessmann et al., 2015; Brown and Mues, 2012). This observation is important for a regime-aware thesis: adding a latent state should be justified by incremental probability quality and economic interpretation, not by an assumption that greater complexity is automatically better.

2.3.2 Origination, Behavioural, and Performance-History Information

Loan-level models separate variables observed at origination from variables updated through servicing. Origination variables describe the initial contract and borrower: credit score, original balance, interest rate, loan term, loan-to-value (LTV), combined LTV, debt-to-income ratio (DTI), occupancy, property type, and number of borrowers. Behavioural variables describe realised performance: current balance, current interest rate, months on book, current delinquency, recent missed payments, cure history, and the recency or frequency of prior delinquency.

This distinction matters because mortgage risk is dynamic. Origination credit score and leverage summarise initial quality, but recent payment behaviour is a direct signal of changing borrower circumstances. A loan that has remained current for several years is different from an otherwise similar loan that recently cured from 60 days past due. Payment-history variables can dominate static characteristics close to default, while origination and macro variables remain important for longer-horizon vulnerability.

Behavioural variables also create a risk of target leakage. Features must be aligned to the prediction month, and no status observed after month t may enter $X_{i,t}$. Similarly, the outcome window must be separated from feature construction. The conceptual requirement in Equation (2.1) is therefore also a data-engineering requirement.

2.3.3 Rare Events and Imbalanced Evaluation

Serious mortgage delinquency is uncommon in many loan-month samples. A classifier that predicts no defaults can achieve high accuracy while being useless. Credit-scoring studies therefore emphasise class-sensitive evaluation, sampling design, and metrics suited to imbalance (Brown and Mues, 2012). ROC-AUC measures ranking across thresholds but can appear strong when the negative class is enormous. Precision-recall curves focus on the positive class and are often more informative for rare events (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).

Neither ROC-AUC nor PR-AUC is sufficient for PD estimation. Both are ranking measures: a monotone transformation of scores can preserve ranking while changing every probability. For expected loss, capital, and portfolio forecasting, the probability scale matters. This motivates the calibration literature reviewed in Section 2.9.

2.4 Economic Mechanisms of Mortgage Default

2.4.1 Equity, Affordability, and Borrower Options

Mortgage default differs from unsecured consumer default because the contract is collateralised and can terminate through competing events such as prepayment, sale, or default. Deng et al. (2000) model termination as the exercise of embedded mortgage options under borrower heterogeneity. The value of the default option increases as property value falls relative to the outstanding balance, while the prepayment option responds to interest-rate movements and refinancing opportunities.

Pure option value does not fully explain observed default. Borrowers with negative equity often continue paying, and borrowers with positive equity can become delinquent following income shocks or household disruption. Elul et al. (2010) frame mortgage default as the interaction of negative equity with liquidity shocks, and Gerardi et al. (2018) distinguish

inability to pay from strategic non-payment by examining unemployment and equity jointly. These studies support a “double-trigger” interpretation: weak equity raises the incentive or inability to resolve the loan through sale, while income or liquidity stress constrains payment.

Theoretical work by Campbell and Cocco (2015) similarly shows that mortgage default reflects house prices, income, inflation, interest rates, and contract characteristics. Payment changes are especially relevant when rates reset or refinancing changes monthly obligations. Tracy and Wright (2016) show that payment reductions can lower expected credit losses, reinforcing the interpretation of payment burden as a causal risk channel rather than merely a correlational feature.

2.4.2 Loan-to-Value and Collateral Dynamics

Original LTV is a standard underwriting variable, but contemporaneous equity depends on both amortisation and house-price changes. A low-LTV loan can become vulnerable after a local price decline, while a high-LTV loan can season into a safer equity position. In large public datasets, current property value is not always observed at the loan level, so models use regional or national house-price indices as proxies. This creates a scale mismatch: macro or regional house-price growth is common to many loans, whereas collateral and local market conditions are heterogeneous.

The mismatch motivates interaction terms. The effect of aggregate house-price deterioration should be stronger for loans with high initial leverage, limited amortisation, or weak borrower liquidity. A purely additive macro term can shift all PDs but cannot represent this heterogeneous amplification unless interactions are included.

2.4.3 Interest Rates, Refinancing, and Monetary Conditions

Interest rates affect mortgage risk through several channels. Contract rates determine scheduled payments; market rates determine refinancing incentives; policy tightening affects borrowing costs and broader demand; and long rates influence both mortgage pricing and asset

valuation. The spread between mortgage rates and government yields also reflects intermediation conditions and market risk premia.

For fixed-rate mortgages, higher market rates do not directly reset the existing contractual payment, but they reduce the ability to refinance or move without losing a favourable rate. For adjustable-rate mortgages, the payment channel is more direct. In either case, rates interact with borrower affordability and loan seasoning. A regime model may summarise the joint state of policy rates, long rates, mortgage rates, and spreads more parsimoniously than a large set of correlated raw covariates.

2.4.4 Labour Income, Unemployment, and Payment History

Employment loss is a central liquidity shock. Aggregate unemployment does not reveal which borrower loses income, but it shifts the distribution of household stress. Initial unemployment claims can provide a faster-moving signal, while payroll and real-income growth describe broader labour and income conditions. These indicators are complementary rather than interchangeable.

The loan-level payment record is the bridge between aggregate labour conditions and realised household strain. Recent 30- or 60-day delinquency, repeated delinquency over rolling windows, time since the last delinquency, and cure behaviour are interpretable state variables. Their marginal effects may change with macro stress. For example, a recent cure during benign labour conditions may have a different prognosis from a cure during widespread deterioration.

2.4.5 Evidence and Relevance Beyond the United States

The empirical model uses Freddie Mac data because it provides a large longitudinal mortgage panel with consistent servicing histories. The research question is nevertheless relevant to emerging and African mortgage markets. Owusu-Manu et al. (2019) examine mortgage default propensity in Ghana and emphasise borrower, institutional, and market determinants.

The exact covariate distributions and mortgage institutions differ from the United States, so model coefficients should not be transported mechanically. The transferable contribution is methodological: a transparent way to combine account-level vulnerability with a public macroeconomic stress signal can be adapted where local performance data become available.

This distinction is particularly important for a journal with a pan-African remit. The study should not imply that a U.S. mortgage model is already validated for African lending. Instead, the Freddie Mac analysis provides a reproducible large-sample test of the regime-aware mechanism, while the literature identifies a future validation agenda involving local underwriting structures, legal environments, currency and inflation dynamics, and data availability.

2.5 Macroeconomic Credit-Risk Models

2.5.1 Observed Macroeconomic Covariates

A macro-augmented loan-level model extends Equation (2.3) as

$$\text{logit}(\pi_{i,t}) = \beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma, \quad (2.4)$$

where M_t contains contemporaneous or lagged macroeconomic variables. Equation (2.4) shifts PD levels when the economy changes. This structure is natural for stress testing because a macroeconomic scenario can be translated into conditional default rates.

Macroeconomic default modelling is well established in corporate and retail credit. Figlewski et al. (2012) link macroeconomic factors to corporate default and rating transitions, while Simons and Rolwes (2009) develop macroeconomic default models for stress testing. In consumer credit, Bellotti and Crook (2009) incorporate macroeconomic variables into survival models, and Bellotti and Crook (2013) use a discrete hazard model to connect retail default rates with stress scenarios. These studies show that economic covariates

matter both for average risk and for the timing of default.

The main modelling choices are variable selection, transformation, lag structure, and forecast information. Levels may capture persistent burden; changes may capture shocks; year-on-year growth may remove seasonality; and spreads can represent relative financing conditions. Macroeconomic series also differ in release timing and revision. A model intended for real-time use should ideally use vintage data or enforce realistic availability lags. The present study uses monthly public series aligned to loan months and treats release-vintage analysis as a further robustness requirement.

2.5.2 Additive Effects Versus Changing Sensitivities

Equation (2.4) assumes that borrower coefficients β are invariant. Yet the economic mechanisms in Section 2.4 imply that the effect of leverage, affordability, or delinquency can depend on the macro state. A fully interacted observed-macro model is

$$\text{logit}(\pi_{i,t}) = \beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma + \sum_{j=1}^p \sum_{\ell=1}^q \theta_{j\ell} X_{i,j,t} M_{\ell,t}. \quad (2.5)$$

Although Equation (2.5) is flexible, it can become high-dimensional and unstable. If p borrower variables interact with q macro variables, the model adds pq terms. Correlated macro indicators complicate interpretation, and a rare-event sample can support spurious interactions despite a large number of rows.

This motivates a lower-dimensional state variable. A latent stress probability can summarise the joint macro environment and serve as a common interaction driver. It reduces the interaction count from pq to approximately p , while preserving a direct economic interpretation: borrower sensitivity changes as the probability of stress rises.

2.5.3 Stress Testing and Temporal Validation

Stress testing asks how risk responds under adverse scenarios, not merely how well a model fits average historical conditions. Ratings and default transitions vary with the business cycle (Bangia et al., 2002). A random train-test split can therefore exaggerate performance because nearby observations share economic conditions and repeated observations from the same loans. Chronological validation is a minimum requirement.

A single terminal holdout remains incomplete. Performance can depend on which episode falls in the test period. Rolling-origin validation estimates the model at several historical cut-offs and evaluates subsequent windows. This design distinguishes a genuine state-dependent relationship from a model that happens to fit one transition. It is especially important for an HMM because state orientation, persistence, and uncertainty can change when the estimation window changes.

2.6 Latent Regimes and Hidden Markov Models

2.6.1 Regime Switching as a Model of Persistence

Regime-switching models formalise the idea that economic observations follow different probability laws across persistent but unobserved states. Hamilton (1989) introduced a widely used Markov-switching approach to business-cycle analysis. Rabiner (1989) provides the canonical HMM formulation and algorithms.

Let $R_t \in \{1, \dots, K\}$ be a latent regime. A first-order homogeneous Markov chain satisfies

$$\Pr(R_t = k \mid R_{t-1} = j, R_{t-2}, \dots) = p_{jk}, \quad \sum_{k=1}^K p_{jk} = 1. \quad (2.6)$$

Conditional on $R_t = k$, the macroeconomic vector Z_t follows a state-dependent emission

distribution. In the Gaussian HMM,

$$Z_t \mid R_t = k \sim \mathcal{N}(\mu_k, \Sigma_k). \quad (2.7)$$

Equations (2.6) and (2.7) separate regime persistence from within-regime observations. The model can represent two months with similar observed conditions but different transition histories, and it produces probabilities rather than deterministic recession labels.

2.6.2 Filtering, Smoothing, and Information Sets

The distinction between filtered and smoothed state probabilities is central to prediction. A filtered probability uses observations through time t :

$$q_t^{\text{filt}} = \Pr(R_t = \text{Stress} \mid Z_{1:t}; \hat{\Theta}), \quad (2.8)$$

whereas a smoothed probability uses the full sequence $Z_{1:T}$:

$$q_t^{\text{smooth}} = \Pr(R_t = \text{Stress} \mid Z_{1:T}; \hat{\Theta}), \quad T > t. \quad (2.9)$$

Equation (2.8) is the appropriate state signal for a real-time PD at t because Equation (2.9) uses future macro observations. A further subtlety is parameter estimation. Even a forward recursion can remain retrospectively informed if $\hat{\Theta}$ and standardisation parameters are estimated using the full sample. A strict real-time design therefore requires expanding-window or rolling estimation of the HMM, not only forward filtering conditional on full-sample parameters.

This distinction is often obscured in empirical applications. Ex-post regime interpretation can legitimately use smoothing, whereas forecasting must enforce the contemporaneous information set. The present empirical pass uses forward probabilities but estimates HMM parameters over the study window; consequently, the resulting signal is best described as

forward-filtered conditional on full-window parameters. Rolling-origin re-estimation is required before a strict real-time claim can be made.

2.6.3 State Identification and Economic Orientation

HMM state labels are arbitrary. Interchanging labels leaves the likelihood unchanged. A stress state must therefore be oriented after estimation using economically justified criteria. With several macro factors, one can define an adverse-direction score that rises with unemployment, rates, claims, and financial tightening and falls with house-price, payroll, income, or sentiment growth. The state with the larger adverse score is labelled stress.

Orientation is not the same as validation. A state can be economically adverse but statistically unhelpful for loan-level PD. The state probability should be inspected for persistence, transition timing, saturation, sensitivity to scaling, sensitivity to initialization, and alignment with known macro episodes. If q_t is nearly zero or one for long intervals, it may act as a coarse time indicator rather than a useful continuous stress measure. If it changes under minor specification choices, downstream interaction estimates will be unstable.

2.6.4 Dimension Reduction and Interpretable Factors

A full-covariance Gaussian HMM becomes fragile as the number of macro variables approaches the number of monthly observations. Correlated series can produce ill-conditioned covariance estimates, while direct use of many variables makes state means difficult to interpret. Dimension reduction is therefore not only computational; it is part of the model's substantive design.

One approach is principal component analysis, which creates orthogonal factors but can mix economic themes. An alternative is transparent grouped factors. Housing/collateral, rate/refinancing, labour/income, and financial-conditions indicators can be oriented so that higher values represent greater stress, standardised, and averaged within groups. This preserves traceability from each factor to its source series. It also prevents a group with many

correlated indicators from dominating merely because it has more variables.

The trade-off is that equal-weight factors impose structure. Factor weights are not estimated to optimise default prediction, and macro series with different signal quality receive similar treatment. This is consistent with the thesis priority of interpretability, but it should be evaluated against simpler raw-variable and alternative-scaling specifications.

2.7 Regime-Switching Models in Credit Risk

Credit-risk applications use regimes at several levels. Transition-matrix models allow rating migrations to change with the business cycle (Bangia et al., 2002). Default-intensity models allow latent states to shift conditional default hazards. Yu et al. (2018) estimate hidden Markov default intensities and show that inferred states can correspond to economic stress. Structural models allow asset dynamics or default boundaries to vary across regimes; Milidonis and Chisholm (2024) report that a regime-switching structural model can respond more rapidly to changes in default risk than a constant-parameter benchmark.

These applications establish that latent states can be useful, but their outputs are not automatically loan-level PDs. A rating migration model describes classes; a corporate intensity model describes firms or portfolios; and a structural model may be designed for traded liabilities. Retail mortgage modelling requires a second layer that combines the common regime with heterogeneous borrower and servicing information.

The proposed bridge is

$$\text{logit}(\pi_{i,t}) = \beta_0 + X_{i,t}^\top \beta + \gamma q_t + q_t X_{i,t}^\top \theta, \quad (2.10)$$

where q_t is the stress probability. The marginal log-odds effect of feature j becomes

$$\frac{\partial \text{logit}(\pi_{i,t})}{\partial X_{i,j,t}} = \beta_j + q_t \theta_j. \quad (2.11)$$

Equation (2.11) gives the model its main interpretation. β_j is the feature effect when $q_t = 0$, and $\beta_j + \theta_j$ is the effect when $q_t = 1$. Intermediate stress probabilities interpolate between them. The interaction vector θ therefore tests whether borrower sensitivities vary with macro stress.

Recent regime-aware consumer-credit work strengthens the motivation but also highlights the scope choice. Brookfield et al. (2026) combine macro sensitivity, regimes, and flexible prediction in a recent preprint. Because it is a preprint and uses a more flexible learning architecture, it is best treated as contemporary motivation rather than settled evidence or a direct benchmark. The present study asks whether the regime mechanism adds value within a transparent logistic structure.

2.8 Machine Learning, Explainability, and Model Governance

Machine-learning credit models can represent nonlinearities and interactions without specifying each term in advance. Khandani et al. (2010) demonstrate the use of machine-learning algorithms in consumer credit risk, and Lessmann et al. (2015) compare a broad set of classification methods. Recent systematic reviews document rapid growth in tree ensembles, support-vector methods, neural networks, hybrid systems, and explainability tools (Shi et al., 2022; Noriega et al., 2023). Mortgage applications increasingly incorporate spatial and temporal structure (Kundig and Sigrist, 2025).

The literature does not imply that the most flexible model is always the best scientific choice. First, classification performance and calibrated probability estimation are distinct. Second, repeated loan-month observations and macro persistence create validation challenges that algorithmic benchmarks can overlook. Third, credit decisions are high stakes. Borrowers, risk managers, auditors, and regulators may require stable reasons for model outputs.

Post-hoc explanation methods can identify local feature contributions or approximate

Table 2.1: Principal model classes in the credit-risk literature and their implications for this study

Model class	Core risk mechanism	Main strength	Limitation for the present question
Structural default	Asset or collateral value relative to debt (Merton, 1974; Campbell and Cocco, 2015)	Strong economic interpretation	Difficult to represent the full servicing history and heterogeneous household shocks
Reduced-form hazard	Conditional default intensity (Lando, 1998; Shumway, 2001)	Natural treatment of timing and censoring	Regime dependence must still be specified
Static logistic scorecard	Borrower and loan covariates (Hand and Henley, 1997; Costa e Silva et al., 2020)	Transparent and operationally familiar	Coefficients are commonly treated as stable through time
Macro-augmented model	Borrower variables plus observed macro covariates (Bellotti and Crook, 2009, 2013)	Connects PD to the credit cycle	Additive effects may not change borrower sensitivities
Latent-regime model	Persistent hidden macro or credit state (Hamilton, 1989; Yu et al., 2018)	Parsimonious nonlinear state representation	Often estimated at corporate, rating, or portfolio level
Machine-learning model	Flexible nonlinear functions and interactions (Khandani et al., 2010; Lessmann et al., 2015)	High predictive capacity	Calibration, stability, and explanation require separate controls

a black-box surface. Bussmann et al. (2021) show how explainable machine learning can support credit-risk management. However, Rudin (2019) argues that high-stakes decisions should prefer interpretable models when such models can perform adequately, rather than treating an explanation of an opaque model as equivalent to transparency.

The regime-aware logistic model is interpretable by construction but not automatically simple. Interactions, standardisation, penalisation, and calibration transform the raw coefficients. Interpretation should therefore report the feature definitions, scaling, selected interactions, regularisation strength, and calibration map. The HMM layer must also be interpretable: state orientation and factor construction should be documented rather than inferred only from a plot.

Machine learning also raises distributional questions. Fuster et al. (2022) show that machine learning can change the distribution of prediction errors and credit-market outcomes across groups. Although the Freddie Mac public sample does not support every fairness analysis, publication should acknowledge that stronger average prediction does not guarantee equitable or transportable decisions. A model trained on U.S. agency mortgage data should not be treated as validated for other borrower populations or institutional settings.

2.9 Calibration and Probability-Forecast Evaluation

2.9.1 Discrimination Is Not Calibration

Discrimination measures whether defaulters tend to receive higher scores than non-defaulters. Calibration measures whether predicted probabilities agree with observed frequencies. A model can have excellent ROC-AUC yet systematically overpredict every PD, because ranking is unchanged by many monotone transformations. Conversely, a well-calibrated but weakly discriminating model may assign nearly the same base-rate probability to every loan.

The distinction is especially important for rare events. Small absolute probability errors can create large portfolio-count errors when multiplied across millions of loan-months.

PD models should therefore report both ranking and probability quality, with calibration evaluated globally and across time or risk segments (Van Calster et al., 2016, 2019).

2.9.2 Proper Scoring Rules

The Brier score for n binary outcomes is

$$\text{BS} = \frac{1}{n} \sum_{r=1}^n (\hat{\pi}_r - y_r)^2. \quad (2.12)$$

The logarithmic score, reported as log loss, is

$$\text{LL} = -\frac{1}{n} \sum_{r=1}^n [y_r \log(\hat{\pi}_r) + (1 - y_r) \log(1 - \hat{\pi}_r)]. \quad (2.13)$$

Both are proper scoring rules: in expectation, the forecaster minimises the score by reporting the true probability (Brier, 1950; Gneiting and Raftery, 2007). Log loss penalises confident errors strongly, while the Brier score has a direct squared-error interpretation and supports decomposition.

For grouped probability bins, the Brier score can be decomposed into reliability, resolution, and uncertainty (Murphy, 1973). Reliability measures disagreement between predicted and observed bin rates; resolution measures how much observed rates vary across bins; and uncertainty depends on the event base rate. The decomposition is diagnostically useful but depends on binning, so it should accompany rather than replace the ungrouped Brier score (Dimitriadis et al., 2020).

2.9.3 Calibration Curves and Calibration Hierarchy

Calibration is not one property. Mean calibration asks whether the average predicted PD equals the event rate. Weak calibration asks whether a recalibration intercept is zero and slope is one. Moderate calibration examines agreement across the prediction range, often

with a smooth calibration curve. Strong calibration would require correct probabilities conditional on every covariate pattern and is generally untestable in finite data (Van Calster et al., 2016).

Reliability diagrams compare predicted and observed rates by bins. They are intuitive but sensitive to bin definitions and sparse tails. Rare-event models should display counts or bin sizes and avoid interpreting noise in extreme bins as structural evidence. Time-slice plots can reveal drift that a pooled curve hides.

2.9.4 Post-Estimation Recalibration

Regularised classifiers can rank well while producing biased probability levels. Recalibration estimates a mapping from raw scores to observed frequencies on a validation set. Platt scaling fits a logistic map to the raw logit or score; intercept-only recalibration adjusts the level while preserving slope; isotonic regression estimates a monotone nonparametric map. Credit-specific work emphasises that PD calibration is part of model construction rather than cosmetic post-processing (Tasche, 2013; Fonseca and Lopes, 2017).

Calibration-method choice must be made without using the test outcomes. A flexible isotonic map can improve in-sample validation fit but be unstable under temporal drift. Platt scaling imposes more structure and may extrapolate more smoothly. Reporting sensitivity across methods is therefore preferable to selecting the method that happens to minimise the terminal test score.

2.9.5 Portfolio Timing and Count Calibration

Loan-level Brier and log-loss averages can conceal economically important monthly errors. For month t , the predicted number of defaults is

$$\hat{D}_t = \sum_{i \in \mathcal{R}_t} \hat{\pi}_{i,t}, \quad (2.14)$$

where $\mathcal{R}_t$ is the risk set, and the observed count is $D_t = \sum_{i \in \mathcal{R}_t} Y_{i,t}$. The error $\hat{D}_t - D_t$ evaluates whether the model tracks portfolio risk through calendar time. A model can slightly improve individual ranking while producing worse portfolio-level timing or level calibration.

Table 2.2: Evaluation dimensions required for a calibration-first regime-aware PD study

Dimension	Representative measure	Question answered	Main caution
Discrimination	ROC-AUC	Are defaults ranked above non-defaults?	Can look strong under severe imbalance
Positive-class retrieval	PR-AUC	Are high-risk predictions concentrated among defaults?	Depends on event prevalence
Probability accuracy	Brier score; log loss	Are probabilities numerically accurate?	Aggregate scores can hide temporal drift
Calibration shape	Reliability curve; intercept; slope	Do predicted and observed rates agree across risk levels?	Bins and smoothers add estimation choices
Portfolio timing	Monthly predicted-default-count error	Does the model track the level of defaults over time?	Large months can dominate absolute errors
Regime robustness	Time slices; stress-probability slices	Does performance persist across macro conditions?	Requires genuine variation in the regime signal

2.10 Critical Synthesis of the Evidence

The literature supports six conclusions. First, loan-level data are necessary because borrower and contract heterogeneity remain central even when common macro conditions matter. Second, mortgage default is generated by interacting equity, affordability, payment-history, and macroeconomic mechanisms. Third, observed macro covariates improve economic responsiveness but an additive structure need not capture changing borrower sensitivities. Fourth, HMMs provide a parsimonious probability of persistent latent stress, but filtering, parameter

estimation, state orientation, and saturation must be handled carefully. Fifth, flexible machine learning is relevant but does not remove the need for calibration, temporal validation, and governance. Sixth, evaluation must combine proper scores, ranking metrics, calibration diagnostics, and portfolio-level timing.

Table 2.3 translates those conclusions into a gap matrix. The final column identifies the specific response implemented or planned in this thesis.

Table 2.3: Literature gap matrix and thesis response

Established evidence	Unresolved issue	Thesis response
Borrower and servicing features predict mortgage default	Static coefficients impose constant sensitivities	Interact selected borrower and payment-history variables with a continuous stress probability
Observed macro variables support stress testing	Many correlated indicators produce unstable or opaque interaction sets	Construct four transparent stress-oriented macro factors
Latent regimes capture persistent economic states	Many studies operate at corporate, ratings, or portfolio scale	Merge a monthly HMM probability into a loan-month logistic PD model
Machine learning captures nonlinearities	Flexibility can weaken transparency and does not ensure calibration	Retain regularised logistic models and use machine learning as related work rather than the main comparator
Calibration can differ from discrimination	Credit studies often foreground AUC or accuracy	Prioritise Brier score, log loss, reliability, calibration sensitivity, and monthly default counts
Chronological holdouts reduce leakage	A single terminal episode may not establish robustness	Use chronological splits now and require rolling-origin HMM and PD re-estimation for publication

The exact contribution can now be stated without overstating novelty. Regime switching is established in macroeconomics and credit risk; logistic PD models are established in consumer lending; and calibration methods are established in probability forecasting. The contribution lies in their disciplined combination for mortgage loan-month data: an interpretable factor-HMM supplies a filtered stress probability; a regularised logistic model uses that probability as both a level effect and an interaction driver; and the resulting PDs are evaluated primarily as probabilities rather than only as rankings.

The literature also sets a high bar for interpreting empirical improvement. A lower terminal test Brier score is not sufficient if validation deteriorates, the regime probability is saturated, or monthly portfolio counts worsen. Evidence for the regime-aware mechanism should ideally satisfy four conditions: improvement over borrower-only and observed-macro baselines; stability across rolling origins; economically coherent interaction effects; and robustness to calibration and HMM specification. These conditions motivate the methodology in Chapter 3 and discipline the interpretation of the results in Chapter 4.

2.11 Research Propositions Derived from the Review

The review yields testable propositions rather than a foregone claim of superiority.

Proposition 1: incremental regime information. If the latent state aggregates macroeconomic information not captured by static borrower variables, then adding q_t should improve out-of-sample probability scores relative to Equation (2.3).

Proposition 2: state-dependent borrower sensitivity. If macro stress changes the consequences of leverage, affordability, or payment distress, then selected components of θ in Equation (2.10) should be economically meaningful and the interaction model should outperform a model containing q_t only as a level effect.

Proposition 3: latent versus observed macro information. If the HMM performs useful dimension reduction, then the clean regime-aware model should remain competitive with or improve on an observed-macro model. If the raw macro model performs better, the latent state may be discarding relevant information.

Proposition 4: calibration stability. If regime awareness captures a persistent risk mechanism rather than a terminal-period coincidence, any improvement should recur across

validation windows, time slices, and calibration methods. Rolling-origin evaluation is the appropriate test.

Proposition 5: portfolio usefulness. If loan-level gains are operationally meaningful, they should not be accompanied by materially worse monthly predicted-default-count errors. Portfolio calibration is therefore a guardrail on individual-level metrics.

These propositions leave open the possibility of a mixed or negative result. That openness is scientifically important. A transparent finding that an HMM regime signal adds little after borrower history and calibration would still clarify the conditions under which regime-aware PD modelling is useful.

Methodology

3.1 Data Sources

The empirical workflow combines two public data sources: Freddie Mac sample mortgage loan files and monthly macroeconomic series from the Federal Reserve Economic Data (FRED) database (Freddie Mac, 2026; Federal Reserve Bank of St. Louis, 2026). The mortgage files provide loan-level origination characteristics and monthly servicing performance records. The macroeconomic data provide the aggregate state variables used to infer the latent regime process.

Table 3.1: Data sources and empirical role

Source	Main variables	Role in the methodology
Freddie Mac sample origination files	Borrower and loan characteristics at origination, including credit quality, leverage, product type, and loan terms.	Define the cross-sectional risk profile of each mortgage.
Freddie Mac sample servicing files	Monthly performance status, delinquency indicators, remaining balance, current activity, and loan age.	Construct the loan-month panel and forward-looking default label.
FRED macroeconomic series	Inflation, policy rates, unemployment, house prices, mortgage rates, Treasury rates, claims, payrolls, real income, financial conditions, and sentiment.	Construct interpretable macro factors, estimate the hidden regime, and produce filtered stress probabilities.

Let $i = 1, \dots, N$ index loans and let $t = 1, \dots, T$ index calendar months. The raw mortgage records are transformed into a panel in which each observation corresponds to a loan-month. The origination file supplies time-invariant or slowly varying underwriting characteristics, while the servicing file supplies monthly performance information. The macroeconomic variables are merged by calendar month so that all loans observed in month t receive the same macro vector.

The raw transformed macro vector is

$$M_t = (M_{t,1}, \dots, M_{t,q})^\top \in \mathbb{R}^q, \quad t = 1, \dots, T. \quad (3.1)$$

Its components are economically meaningful levels, changes, spreads, or growth rates. The candidate set comprises CPI inflation, the Federal Funds rate, unemployment and its change, house-price growth, the 30-year mortgage rate, the mortgage–Treasury spread, the 10-year Treasury rate and its change, the 10-year minus 2-year yield spread, initial claims and their change, payroll growth, real disposable-income growth, the National Financial Conditions Index, consumer sentiment, and sentiment change. Weekly series are aggregated to monthly frequency before the merge.

To avoid fitting a high-dimensional full-covariance HMM directly, the variables are oriented so that larger values represent greater stress and grouped into four transparent factors: housing/collateral, rate/refinancing, labour/income, and financial conditions. If $\mathcal{G}_g$ is the set of source variables assigned to group g and $a_j \in \{-1, 1\}$ is the adverse-direction sign, then

$$F_{t,g} = \frac{1}{|\mathcal{G}_g|} \sum_{j \in \mathcal{G}_g} \frac{a_j M_{t,j} - \overline{a_j M_j}}{s(a_j M_j)}, \quad g = 1, \dots, r. \quad (3.2)$$

Equal weighting preserves a clear mapping from source indicators to HMM inputs and prevents a group with many correlated series from dominating by construction. The factor vector $F_t = (F_{t,1}, \dots, F_{t,r})^\top$, with $r = 4$ when all groups are available, is standardised again

before HMM estimation:

$$Z_{t,g} = \frac{F_{t,g} - \bar{F}_g}{s_g}, \quad g = 1, \dots, r. \quad (3.3)$$

The current expanded empirical pass estimates these quantities and the HMM parameters over January 2014–September 2024, then computes the state recursion forward through that window. The reported state probabilities are therefore filtered with respect to observations but conditional on full-window transformation and HMM parameters. A strict real-time robustness design must re-estimate the factors, scaler, and HMM at each rolling origin; the manuscript retains this distinction explicitly rather than claiming that the present HMM is fully out of sample.

3.2 Unit Of Analysis

The unit of observation is one mortgage at one month. For loan i in month t , the full observation is written as

$$\mathcal{O}_{i,t} = \left(Y_{i,t}^{(12)}, X_{i,t}, M_t, \hat{p}_t^{\text{Stress}} \right), \quad (3.4)$$

where $Y_{i,t}^{(12)}$ is the 12-month serious-delinquency label, $X_{i,t}$ is the borrower and loan feature vector, M_t is the observed macro vector, and $\hat{p}_t^{\text{Stress}}$ is the filtered probability that month t is in the stress regime.

The borrower and loan feature vector is represented as

$$X_{i,t} = (X_{i,t,1}, X_{i,t,2}, \dots, X_{i,t,p})^\top \in \mathbb{R}^p. \quad (3.5)$$

The intercept is modelled separately in the logistic equations. The components of $X_{i,t}$ consist of origination variables and current performance variables observed at month t . The feature list is fixed before model estimation and excludes any variable measured after month t . The construction follows the information principle that only information available at or before

month t may be used to predict the event over months $t + 1$ through $t + 12$.

The empirical panel is defined over the at-risk population. A loan-month is at risk when the loan is active and not already seriously delinquent at month t . Let $D_{i,t}$ denote delinquency at month t , $T_i^{\max}$ the last observed month for loan i , and T_i^{term} its first terminal-servicing month, if any. The implemented follow-up indicator is

$$W_{i,t}^{(12)} = \mathbf{1}\{t + 12 \leq T_i^{\max}\} \mathbf{1}\{T_i^{\text{term}} \text{ is missing or } T_i^{\text{term}} > t + 12\}. \quad (3.6)$$

Let $\mathcal{A}_{i,t}$ denote the event that the loan is active and eligible for forward prediction. The modelling sample is restricted to

$$\mathcal{R} = \left\{ (i, t) : \mathcal{A}_{i,t} = 1, D_{i,t} < 90, W_{i,t}^{(12)} = 1 \right\}. \quad (3.7)$$

This restriction requires the observed servicing history to extend beyond the twelve-month horizon and excludes loans terminating inside that horizon. It prevents the model from learning from observations already in the target state and avoids treating mechanically truncated follow-up as a non-event.

3.3 Primary Label

The primary response variable is a 12-month transition to serious delinquency, defined as reaching 90 or more days past due within the next 12 months. For each at-risk loan-month $(i, t) \in \mathcal{R}$, the label is

$$Y_{i,t}^{(12)} = \mathbf{1} \left\{ \max_{h=1,\dots,12} D_{i,t+h} \geq 90 \right\}. \quad (3.8)$$

This definition aligns the statistical target with a forward-looking PD concept: the question is not whether the loan is currently distressed, but whether it transitions into serious delinquency over a fixed prediction horizon.

The conditional probability of interest is

$$\pi_{i,t} = \Pr \left(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t \right), \quad (3.9)$$

where $\mathcal{F}_t$ denotes the information available at month t . In the borrower-only baseline, $\mathcal{F}_t$ contains loan-level information only. In the macro-augmented baseline, it also contains observed macro variables. In the regime-aware model, it contains the filtered stress probability inferred from the macro HMM.

The 90+ days-past-due threshold is used because serious delinquency is a standard operational default proxy in credit risk and is closely related to regulatory default definitions based on material arrears (Basel Committee on Banking Supervision, 2004; Board of Governors of the Federal Reserve System, 2006). The fixed 12-month horizon creates a consistent target across the panel and supports direct comparison of predicted probabilities with realised default frequencies.

3.4 Macro Regime Inference

The latent macro regime is modelled using a two-state Gaussian hidden Markov model. Let

$$R_t \in \{1, 2\} \quad (3.10)$$

denote the unobserved macroeconomic regime in month t . After estimation, the two states are interpreted economically and labelled as Expansion and Stress according to their fitted macro profiles.

The regime process is assumed to follow a first-order homogeneous Markov chain:

$$\Pr(R_t = k \mid R_{t-1} = j, R_{t-2}, \dots, Z_{1:t-1}) = \Pr(R_t = k \mid R_{t-1} = j) = p_{jk}, \quad (3.11)$$

for $j, k \in \{1, 2\}$. The transition matrix is

$$P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}, \quad p_{jk} \geq 0, \quad \sum_{k=1}^2 p_{jk} = 1. \quad (3.12)$$

The diagonal entries p_{11} and p_{22} measure regime persistence. The off-diagonal entries measure transition probabilities between expansion and stress.

Conditional on the latent regime, the standardised macro vector follows a multivariate Gaussian distribution:

$$Z_t \mid R_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad k = 1, 2. \quad (3.13)$$

The complete HMM parameter vector is

$$\Theta = (\delta, P, \mu_1, \mu_2, \Sigma_1, \Sigma_2), \quad (3.14)$$

where $\delta = (\delta_1, \delta_2)$ is the initial regime distribution.

For an observed factor path $Z_{1:T}$ and a latent regime path $R_{1:T}$, the joint density is

$$f(Z_{1:T}, R_{1:T}; \Theta) = \delta_{R_1} \left[\prod_{t=2}^T p_{R_{t-1}, R_t} \right] \left[\prod_{t=1}^T \phi_r(Z_t; \mu_{R_t}, \Sigma_{R_t}) \right], \quad (3.15)$$

where $\phi_r(z; \mu, \Sigma)$ denotes the r -dimensional Gaussian density:

$$\phi_r(z; \mu, \Sigma) = \frac{1}{(2\pi)^{r/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (z - \mu)^\top \Sigma^{-1} (z - \mu) \right\}, \quad (3.16)$$

with $\mu \in \mathbb{R}^r$ and positive-definite $\Sigma \in \mathbb{R}^{r \times r}$.

Because the regime path is unobserved, direct maximisation of the complete-data likelihood is not feasible. The model is estimated with the Baum-Welch algorithm, an expectation-maximisation procedure for HMMs (Baum et al., 1970; Bilmes, 1998). In the E-step at

iteration q , the smoothed state probabilities are computed:

$$\gamma_t(k) = \Pr(R_t = k \mid Z_{1:T}; \Theta^{(q)}), \quad (3.17)$$

and the smoothed transition probabilities are

$$\xi_t(j, k) = \Pr(R_{t-1} = j, R_t = k \mid Z_{1:T}; \Theta^{(q)}), \quad t = 2, \dots, T. \quad (3.18)$$

In the M-step, the parameters are updated by weighted averages:

$$\delta_k^{(q+1)} = \gamma_1(k), \quad (3.19)$$

$$p_{jk}^{(q+1)} = \frac{\sum_{t=2}^T \xi_t(j, k)}{\sum_{t=2}^T \sum_{\ell=1}^2 \xi_t(j, \ell)}, \quad (3.20)$$

$$\mu_k^{(q+1)} = \frac{\sum_{t=1}^T \gamma_t(k) Z_t}{\sum_{t=1}^T \gamma_t(k)}, \quad (3.21)$$

$$\Sigma_k^{(q+1)} = \frac{\sum_{t=1}^T \gamma_t(k) (Z_t - \mu_k^{(q+1)}) (Z_t - \mu_k^{(q+1)})^\top}{\sum_{t=1}^T \gamma_t(k)}. \quad (3.22)$$

The E-step and M-step are iterated until the observed-data log-likelihood stabilises:

$$\ell(\Theta) = \log \left[\sum_{R_1=1}^2 \cdots \sum_{R_T=1}^2 f(Z_{1:T}, R_{1:T}; \Theta) \right]. \quad (3.23)$$

For downstream PD prediction, the relevant macro signal is not the smoothed probability in Equation (3.17). Smoothed probabilities condition on the full factor path $Z_{1:T}$ and therefore use future observations. The PD model instead uses the forward-filtered probability conditional on the fitted full-window parameters:

$$\alpha_t(k) = f(Z_{1:t}, R_t = k; \hat{\Theta}), \quad (3.24)$$

with recursion

$$\alpha_1(k) = \hat{\delta}_k \phi_r(Z_1; \hat{\mu}_k, \hat{\Sigma}_k), \quad (3.25)$$

$$\alpha_t(k) = \phi_r(Z_t; \hat{\mu}_k, \hat{\Sigma}_k) \sum_{j=1}^2 \alpha_{t-1}(j) \hat{p}_{jk}, \quad t = 2, \dots, T. \quad (3.26)$$

$$\hat{p}_t(k) = \Pr(R_t = k \mid Z_{1:t}; \hat{\Theta}) = \frac{\alpha_t(k)}{\sum_{\ell=1}^2 \alpha_t(\ell)}. \quad (3.27)$$

The stress input used in the credit model is

$$\hat{p}_t^{\text{Stress}} = \hat{p}_t(k^*), \quad k^* = \arg \max_{k \in \{1,2\}} s(k), \quad (3.28)$$

where $s(k)$ is an economic stress score assigned from the fitted factor profile. Since every factor is oriented so that larger values indicate greater stress pressure, the study uses

$$s(k) = \sum_{g=1}^r \mu_{k,g}, \quad k = 1, 2. \quad (3.29)$$

Thus, the stress state is the fitted state with the larger combined mean across the adverse-oriented factors.

3.5 Model Comparison

The thesis centres on three logistic specifications: a borrower-only baseline, a macro-augmented baseline, and a clean regime-aware model. The implemented nested ladder adds three diagnostic specifications so the incremental value of the stress level, stress interactions, raw macro variables, and stress dynamics can be separated while preserving a common sample and target.

Table 3.2: Model hierarchy used in the empirical comparison

Model	Specification	Purpose
Borrower-only logit	Borrower and loan features $X_{i,t}$ only.	Establish the static loan-level benchmark.
Macro-augmented logit	Borrower features plus transformed raw macro variables M_t .	Test whether observed macro variables improve PD estimation additively.
Baseline plus stress	Borrower features plus q_t without interactions.	Isolate the direct level information in the latent state.
Regime-aware logit	Borrower features, filtered stress probability, and stress-by-borrower interactions.	Test whether latent stress changes borrower risk sensitivities.
Regime-aware plus raw macro	Clean regime-aware specification plus M_t .	Test whether the HMM signal adds information beyond observed macro covariates.
Dynamic-stress robustness	Clean regime-aware specification plus Δq_t and stress duration.	Test a small dynamic extension without redefining the core model.

The borrower-only baseline is

$$\pi_{i,t}^{(0)} = \Pr(Y_{i,t}^{(12)} = 1 \mid X_{i,t}) = \sigma(\beta_0 + X_{i,t}^\top \beta), \quad (3.30)$$

where $\sigma(a) = (1 + \exp(-a))^{-1}$. This model assumes that borrower and loan characteristics have fixed effects through time.

The macro-augmented baseline is

$$\pi_{i,t}^{(M)} = \sigma(\beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma_M). \quad (3.31)$$

This model allows the macroeconomic environment to shift the PD, but it does not allow the effect of borrower variables to change with the macro state.

The proposed regime-aware model is

$$\pi_{i,t}^{(R)} = \sigma(\beta_0 + X_{i,t}^\top \beta + \gamma_R \hat{p}_t^{\text{Stress}} + \hat{p}_t^{\text{Stress}} X_{i,t}^\top \theta). \quad (3.32)$$

The coefficient vector β measures baseline borrower effects. The scalar γ_R measures the direct level effect of macro stress probability. The interaction vector θ measures how borrower sensitivities change as the economy moves toward the stress regime.

To see the interpretation of the interaction terms, consider a continuous borrower feature $X_{i,t,j}$. The marginal effect on the log-odds is

$$\frac{\partial}{\partial X_{i,t,j}} \text{logit}(\pi_{i,t}^{(R)}) = \beta_j + \hat{p}_t^{\text{Stress}} \theta_j. \quad (3.33)$$

When $\theta_j > 0$, the log-odds contribution of feature j becomes stronger in stress. When $\theta_j < 0$, the contribution weakens as stress probability rises. For binary or categorical indicators, the same expression is interpreted as a stress-dependent discrete log-odds difference rather than a derivative. This is the main mathematical distinction between the regime-aware model

and the macro-augmented baseline.

The dynamic robustness variant adds

$$\Delta q_t = q_t - q_{t-1}, \quad D_t^{\text{Stress}} = \begin{cases} D_{t-1}^{\text{Stress}} + 1, & q_t \geq 0.5, \\ 0, & q_t < 0.5, \end{cases} \quad (3.34)$$

to the clean regime-aware model. Lagged probabilities at 3, 6, and 12 months are generated for diagnostics, but the reported smaller robustness model uses only Δq_t and D_t^{Stress} .

The logistic coefficients are estimated by penalised maximum likelihood. For a generic model with predicted probabilities $\hat{\pi}_{i,t}$, the Bernoulli log-likelihood over the training set $\mathcal{T}$ is

$$\ell = \sum_{(i,t) \in \mathcal{T}} \left[Y_{i,t}^{(12)} \log(\hat{\pi}_{i,t}) + (1 - Y_{i,t}^{(12)}) \log(1 - \hat{\pi}_{i,t}) \right]. \quad (3.35)$$

All numerical features and constructed interactions are standardised inside the pipeline. The empirical implementation uses stochastic-gradient logistic regression with an L_2 penalty on the complete non-intercept coefficient vector:

$$\hat{\Omega} = \arg \min_{\Omega} \{ -\ell(\Omega) + \lambda \|\omega\|_2^2 \}, \quad \lambda \geq 0, \quad (3.36)$$

where $\Omega = (\beta_0, \omega)$ and ω contains every non-intercept coefficient in the relevant specification. The penalty stabilises correlated macro and interaction features (le Cessie and van Houwelingen, 1992). The implementation evaluates $\lambda \in \{3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}\}$ on deterministic 500,000-row training and validation subsamples, selects the value with the lowest validation Brier score, and then fits the selected pipeline on the full training sample.

3.6 Evaluation Plan

All comparisons use chronological splits. This is necessary because the task is forward-looking: a credit model trained with future information would overstate performance. Let

$\tau_1 < \tau_2$ denote calendar-month cutoffs. The full modelling sample is partitioned into training, validation, and test periods as

$$\begin{aligned}\mathcal{R}_{\text{train}} &= \{(i, t) \in \mathcal{R} : t \leq \tau_1\}, \\ \mathcal{R}_{\text{valid}} &= \{(i, t) \in \mathcal{R} : \tau_1 < t \leq \tau_2\}, \\ \mathcal{R}_{\text{test}} &= \{(i, t) \in \mathcal{R} : t > \tau_2\}.\end{aligned}\tag{3.37}$$

The current empirical pass estimates one HMM over the mortgage study window plus a 12-month lookback. The forward recursion ensures that the state update at t does not use factor observations after t , but its fitted scaler and HMM parameters use the full study window. This is less forward-looking than smoothing the state sequence, but it is not a strict real-time design. The required publication robustness pass will re-estimate the factor transforms and HMM on expanding historical windows. The downstream PD models are fit on $\mathcal{R}_{\text{train}}$, tuned on $\mathcal{R}_{\text{valid}}$, calibrated on validation predictions, and evaluated once on $\mathcal{R}_{\text{test}}$ in the present results.

The primary evaluation criterion is calibration. The Brier score is

$$BS = \frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} \left(\hat{\pi}_{i,t} - Y_{i,t}^{(12)} \right)^2, \tag{3.38}$$

where $\mathcal{S}$ denotes the evaluation sample and $|\mathcal{S}|$ is its number of loan-month observations. Lower values indicate better probabilistic accuracy (Brier, 1950).

The Brier score can be decomposed into reliability, resolution, and uncertainty. For binned calibration diagnostics, let predictions be grouped into B bins, with n_b observations in bin b , $n = |\mathcal{S}|$, average predicted probability $\bar{\pi}_b$, observed default rate $\bar{y}_b$, and overall default rate $\bar{y}$. The grouped Murphy decomposition is

$$BS_{\text{grouped}} = \underbrace{\sum_{b=1}^B \frac{n_b}{n} (\bar{\pi}_b - \bar{y}_b)^2}_{\text{Reliability}} - \underbrace{\sum_{b=1}^B \frac{n_b}{n} (\bar{y}_b - \bar{y})^2}_{\text{Resolution}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{Uncertainty}}. \tag{3.39}$$

Reliability measures calibration error. Resolution measures the ability of the model to separate groups with different observed default frequencies. Uncertainty depends only on the base default rate (Murphy, 1973; Dimitriadis et al., 2020). The grouped decomposition is a diagnostic summary; the ungrouped Brier score in Equation 3.38 remains the primary reported score.

Calibration is also evaluated with bin-level observed-versus-predicted default rates:

$$\widehat{\text{Cal}}_b = \left(\frac{1}{n_b} \sum_{(i,t) \in b} \hat{\pi}_{i,t}, \frac{1}{n_b} \sum_{(i,t) \in b} Y_{i,t}^{(12)} \right), \quad b = 1, \dots, B. \quad (3.40)$$

A well-calibrated model should produce bin-level points with small vertical deviations from the 45-degree line; the reliability term in Equation 3.39 summarises these deviations numerically. Calibration is examined overall, by calendar period, and by regime slice. The regime-slice check is central to the thesis because the proposed model is intended to improve probability quality during macro stress and transition periods.

Three validation-fitted calibration maps are compared. Platt scaling fits

$$\hat{\pi}^{\text{Platt}} = \sigma(a + b \logit(\hat{\pi}^{\text{raw}})), \quad (3.41)$$

intercept-only recalibration fixes $b = 1$, and isotonic regression estimates a non-decreasing map from raw to calibrated probabilities. The primary output uses Platt scaling for continuity across specifications; the other maps are reported as sensitivity analysis. Calibration parameters are estimated only from the chronological validation predictions.

Portfolio-level timing is assessed by the monthly predicted-default count

$$\hat{D}_t = \sum_{i \in \mathcal{R}_t} \hat{\pi}_{i,t}, \quad D_t = \sum_{i \in \mathcal{R}_t} Y_{i,t}^{(12)}, \quad (3.42)$$

with absolute error $|\hat{D}_t - D_t|$ and absolute relative error $|\hat{D}_t - D_t|/D_t$. This diagnostic tests whether improvements in average loan-level scores translate into useful portfolio risk levels

through time.

Secondary metrics include log loss, ROC-AUC, and PR-AUC. Log loss is

$$LL = -\frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} \left[Y_{i,t}^{(12)} \log(\hat{\pi}_{i,t}) + (1 - Y_{i,t}^{(12)}) \log(1 - \hat{\pi}_{i,t}) \right]. \quad (3.43)$$

For numerical stability, predicted probabilities are clipped to $[\varepsilon, 1 - \varepsilon]$ before evaluating the logarithms, with $\varepsilon = 10^{-15}$ fixed before testing. ROC-AUC is retained as a ranking measure, while PR-AUC is reported because serious delinquency is rare and precision-recall analysis is informative for imbalanced binary outcomes (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015).

The empirical interpretation is based on the full pattern of evidence rather than one metric alone. The regime-aware model is considered useful only if probability-score gains are accompanied by credible temporal, calibration, and portfolio diagnostics. Any deterioration relative to the baselines is reported explicitly. This criterion follows the purpose of the model: to produce interpretable and macro-sensitive PD estimates, not merely to maximise a black-box accuracy measure.

Table 3.3: Evaluation metrics and interpretation

Metric	Main question answered	Role in this thesis
Brier score	What is the mean squared error of the predicted PDs?	Primary probability-quality metric.
Reliability diagram	Do predicted PD bands match observed default rates?	Main visual calibration diagnostic.
Brier decomposition	Are changes driven by reliability, resolution, or base-rate uncertainty?	Explains the source of model improvement.
Log loss	Does the model assign high probability to realised outcomes?	Secondary proper scoring rule.
ROC-AUC	Does the model rank defaulters above non-defaulters?	Secondary discrimination metric.
PR-AUC	How well does the model identify rare default events?	Secondary rare-event metric.
Monthly count error	Does the sum of predicted PDs track realised defaults by month?	Portfolio-level calibration guardrail.

Results

4.1 Empirical Coverage

The expanded empirical run uses Freddie Mac sample cohorts from 2015 through 2024 after enforcing the twelve-month look-ahead requirement. Although 2025 source files are included in ingestion, that cohort cannot contribute model-ready observations because twelve months of subsequent servicing performance are not yet available. The final model-ready panel contains 15,461,471 loan-month observations and 167,056 forward transitions to 90+ days past due, an overall event rate of 1.08%.

The chronological split produces 8,179,648 training observations, 3,175,548 validation observations, and 4,106,275 test observations. The corresponding event rates are 1.519%, 0.498%, and 0.657%. The decline from training to validation and the subsequent rise in the test period demonstrate why a random split would be inappropriate: base risk and the borrower mix change materially through calendar time.

Coverage is not uniform across cohorts. Earlier cohorts contribute longer performance histories, while 2023 and 2024 contribute fewer months because of the common September 2024 model-ready endpoint. The analysis therefore treats loan-months, not origination cohorts, as the prediction units and reports calendar-time diagnostics in addition to pooled scores.

4.2 Expanded Macroeconomic Regime Layer

The macro ingestion contains 862 monthly rows from July 1954 through May 2026, but the active HMM estimation window is restricted to January 2014–September 2024. The transformed candidate set is reduced to four adverse-oriented factors: housing/collateral, rate/refinancing, labour/income, and financial conditions. The two-state Gaussian HMM is fitted to those factors, and the state with the larger sum of adverse-oriented means is labelled stress.

Figure 4.1 displays the resulting stress probability. The path is informative in the earlier study window but becomes nearly degenerate in the terminal period. Across the test months, q_t ranges from approximately 0.999903 to a value numerically indistinguishable from 1. Every test observation is consequently classified as hard stress under the threshold $q_t \geq 0.5$.

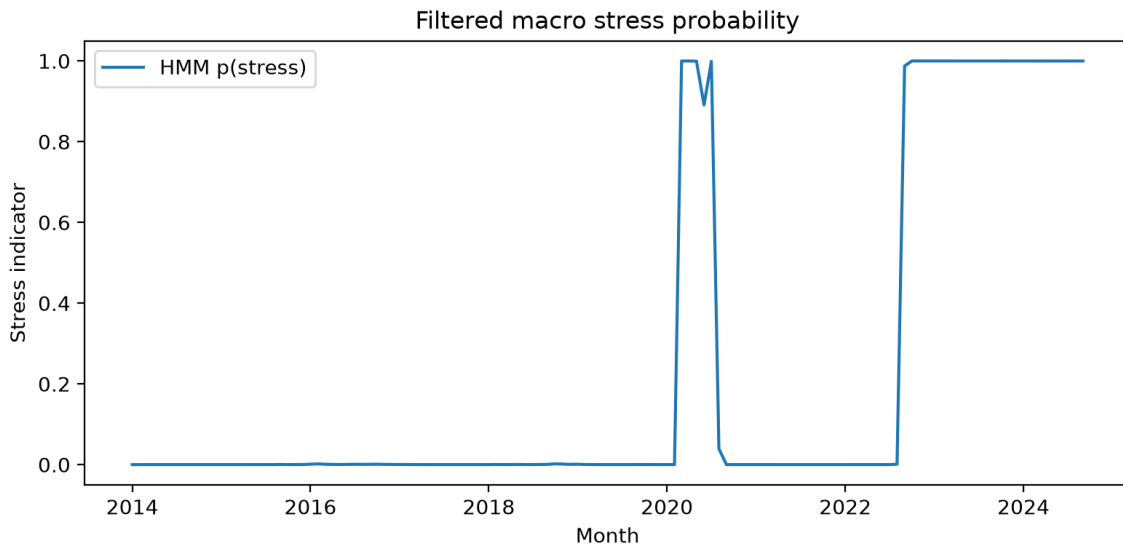


Figure 4.1: Forward-filtered probability of the HMM stress state over the mortgage study window. The probabilities are conditional on factor transformations and HMM parameters estimated over the full displayed window.

This saturation has two consequences. First, a hard normal-versus-stress test comparison is impossible because the test set contains no hard-normal months. Second, even decile labels provide little economic separation: the deciles partition extremely small numerical differences close to one. The stress-bin results are retained as diagnostics but cannot establish that the

model performs differently across substantively distinct test regimes.

4.3 Validation Evidence and Model Selection

Regularisation is selected from $\{3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}\}$ using Brier score on deterministic 500,000-row training and validation subsamples. The borrower-only, macro, stress-only, regime-aware-with-raw-macro, and dynamic-stress specifications select 3×10^{-5} , while the clean regime-aware model selects 3×10^{-6} . The fitted probabilities are then recalibrated on the full chronological validation set.

Validation evidence does not favour the clean regime-aware model. Under primary Platt calibration, the borrower-only model records a validation Brier score of 0.004645 and log loss of 0.026122. The clean regime-aware model records 0.004682 and 0.027021. The borrower-only model also has the strongest validation ROC-AUC (0.798634) and PR-AUC (0.114809) among the reported ladder. Thus, the eventual test advantage of the regime-aware model was not selected by looking at the test set and is not a uniformly recurring advantage across time.

Isotonic calibration improves validation Brier scores for several models. For the borrower-only model, isotonic Brier is 0.004629, while the clean regime-aware isotonic Brier is 0.004647. These results support reporting calibration sensitivity but do not change the validation ranking.

4.4 Held-Out Test Comparison

Table 4.1 reports the primary Platt-calibrated test results. The clean regime-aware model has the lowest Brier score and log loss and the highest ROC-AUC and PR-AUC. Relative to borrower-only, its Brier score decreases from 0.006055 to 0.006015, a reduction of approximately 0.66%, and log loss decreases by approximately 0.49%. ROC-AUC increases by 0.00528 and PR-AUC by 0.01174.

Table 4.1: Held-out test performance using the primary Platt-calibrated probabilities

Model	Mean PD	Brier	Log loss	ROC-AUC	PR-AUC
Borrower-only	0.010747	0.006055	0.033106	0.821979	0.148361
Observed macro	0.009743	0.006078	0.033453	0.805984	0.133367
Baseline plus stress	0.013348	0.006089	0.034349	0.825747	0.150199
Clean regime-aware	0.010802	0.006015	0.032945	0.827256	0.160097
Regime-aware plus raw macro	0.011313	0.006105	0.034187	0.804293	0.131448
Dynamic-stress robustness	0.000687	0.006472	0.045723	0.745641	0.077918

Note: $n = 4,106,275$ loan-months, with 26,983 events and an event rate of 0.006571. Lower Brier and log loss are better; higher ROC-AUC and PR-AUC are better. Source: author's computation from the active empirical pipeline.

The nested ladder clarifies what produces the improvement. Adding q_t without interactions does not improve probability quality: its Brier score is 0.006089. Adding observed macro variables to the interaction model also performs worse than the clean regime-aware specification. The evidence therefore favours the selected stress-by-borrower interaction structure on this test period, not the claim that every use of a macro regime improves PD estimation.

The dynamic-stress robustness model performs poorly. Its mean predicted PD is 0.000687 against an event rate of 0.006571, indicating severe underprediction after validation calibration. Its ranking metrics also deteriorate substantially. Lagged or duration effects should therefore not replace the clean model in the main specification without redesign and rolling-origin testing.

4.5 Calibration Results

Figure 4.2 displays the test reliability curves generated by the empirical pipeline. The curves show that validation-fitted calibration does not fully eliminate temporal drift. Mean predicted PD exceeds the test event rate for each core model: 0.010747 for borrower-only and 0.010802 for regime-aware against 0.006571 observed. The regime-aware advantage is therefore a relative improvement in distribution-sensitive probability scores, not exact mean

calibration.

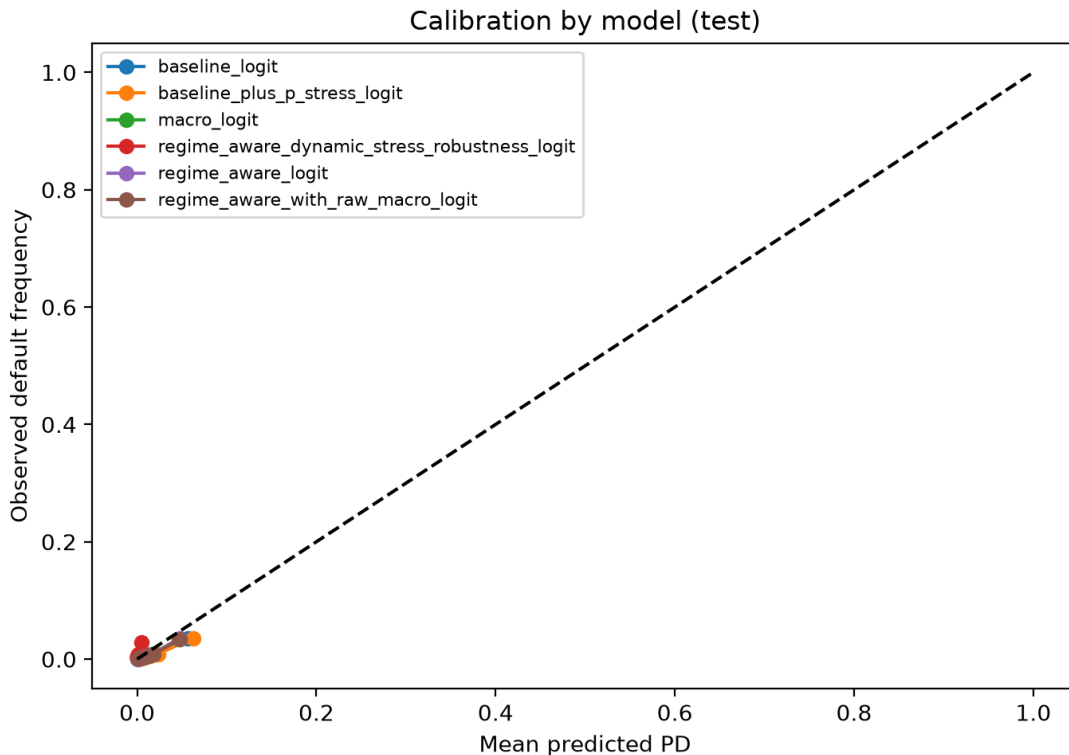


Figure 4.2: Test-period calibration curves for the empirical model ladder. Points compare mean predicted PD with observed default frequency in prediction bins.

The grouped Brier decomposition provides additional context. The clean regime-aware model has reliability 0.0000293 and resolution 0.0001017. Borrower-only has reliability 0.0000549 and resolution 0.0001006. On the grouped diagnostic, the regime-aware improvement is therefore primarily a lower reliability error while resolution remains similar. The observed-macro model has even lower grouped reliability (0.0000247), but its ungrouped Brier and ranking metrics are worse. This difference illustrates why binned decomposition should not be used alone.

Table 4.2 compares calibration methods for the two central models. Isotonic calibration slightly improves the clean regime-aware test Brier and log loss relative to Platt, but it does not improve borrower-only performance on the test set. Because calibration methods were fitted on validation and the primary method was fixed as Platt, isotonic is reported as

sensitivity rather than promoted after observing the test result.

Table 4.2: Calibration-method sensitivity for the principal test models

Model	Calibration	Mean PD	Brier	Log loss
Borrower-only	Raw	0.035379	0.008941	0.052552
Borrower-only	Intercept-only	0.011309	0.006120	0.033404
Borrower-only	Isotonic	0.011024	0.006125	0.033322
Borrower-only	Platt (primary)	0.010747	0.006055	0.033106
Clean regime-aware	Raw	0.063093	0.015897	0.084550
Clean regime-aware	Intercept-only	0.013791	0.006673	0.035555
Clean regime-aware	Isotonic	0.011020	0.006007	0.032811
Clean regime-aware	Platt (primary)	0.010802	0.006015	0.032945

Note: Calibration maps are estimated on the chronological validation set and applied unchanged to the test set. Source: author’s computation.

4.6 Temporal and Portfolio-Level Diagnostics

The clean regime-aware model has a lower Brier score in both reported test-year slices. In 2023, its Brier score is 0.005432 compared with 0.005445 for borrower-only. In 2024, the values are 0.006546 and 0.006609. This consistency across the two terminal years is encouraging, although the slices share the same fitted HMM and both are classified as hard stress.

Portfolio timing is less favourable. Table 4.3 reports the mean absolute monthly predicted-default-count error over 18 test months. Borrower-only has the lowest error at 952.5 defaults per month, compared with 965.3 for the clean regime-aware model. The mean absolute relative errors are 60.97% and 63.67%. Both are large, and the errors increase toward the end of the test period as predicted PDs rise faster than observed event rates.

The portfolio result tempers the pooled test advantage. The clean regime-aware model improves average loan-level probability scores and ranking but does not improve the average monthly level of predicted defaults. This distinction is material for a model intended to support portfolio planning.

Table 4.3: Monthly predicted-default-count error over the 18-month test period

Model	Mean absolute count error	Mean absolute relative error
Borrower-only	952.5	0.6097
Clean regime-aware	965.3	0.6367
Observed macro	1212.3	0.9016
Dynamic-stress robustness	1342.3	0.8751
Regime-aware plus raw macro	1347.5	0.9957
Baseline plus stress	1546.0	1.0075

Note: Absolute relative error is computed month by month relative to the observed number of defaults and then averaged. Source: author’s computation.

4.7 Discrimination Diagnostics

Figure 4.3 shows the test ROC curves. The clean regime-aware model improves ROC-AUC relative to borrower-only, while models containing the full raw macro vector rank worse. PR-AUC yields the same ordering for the clean model and borrower-only. Because the event rate is below one percent, PR-AUC remains the more demanding summary of positive-class retrieval.

4.8 Interpretation and Unresolved Robustness Requirements

The current evidence supports a narrow claim: on the 2023–2024 terminal test period, the clean regime-aware interaction model modestly improves Brier score, log loss, ROC-AUC, and PR-AUC relative to borrower-only. It does not support a broad claim of dominance. Borrower-only is stronger on validation Brier and monthly default-count error; the direct stress-only, raw-macro interaction, and dynamic-stress variants do not improve probability quality; and the test-period stress signal is nearly saturated.

Three robustness requirements remain before journal submission. First, the factor transforms, scaler, and HMM must be re-estimated at multiple historical origins so that the regime probability is genuinely available in real time. Second, uncertainty around paired

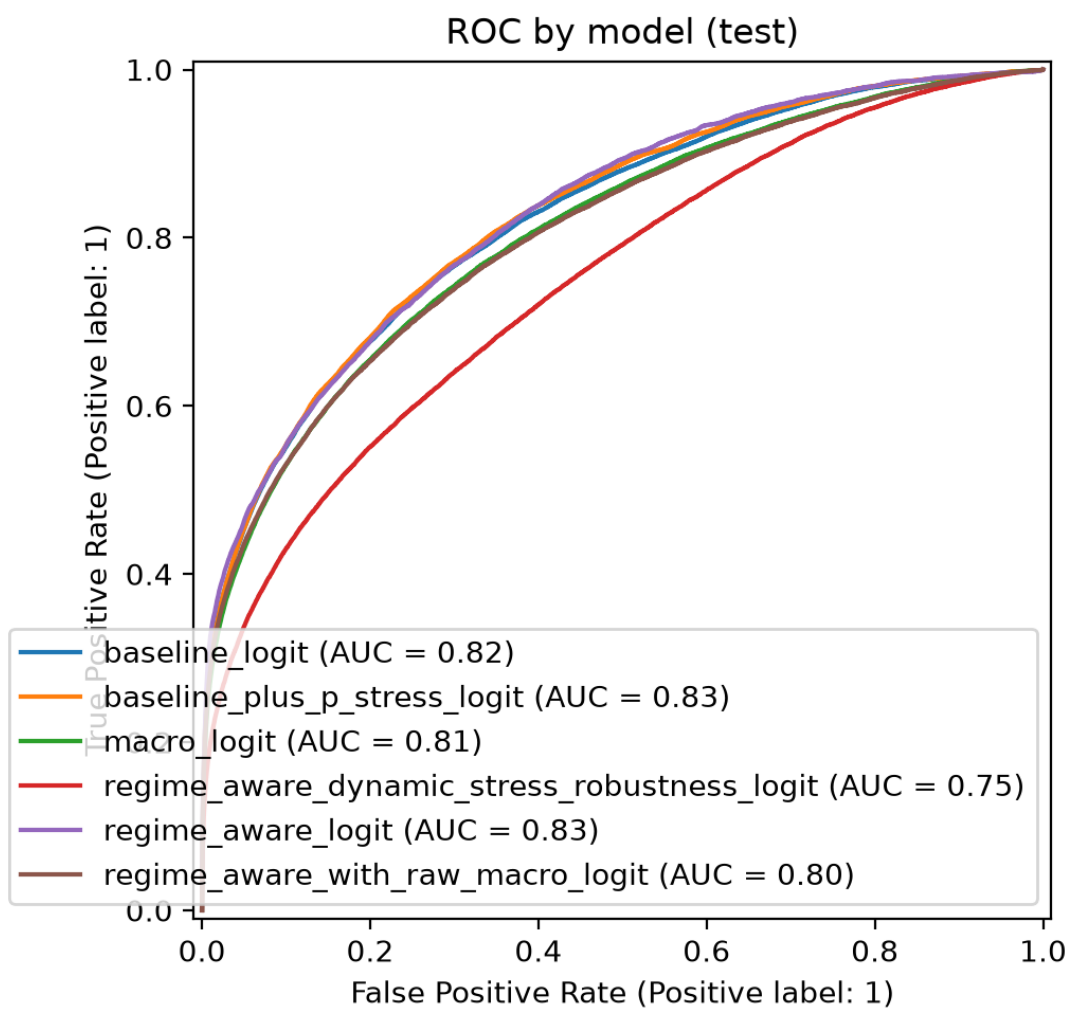


Figure 4.3: Receiver operating characteristic curves for the held-out test period. ROC-AUC is reported as a secondary discrimination measure rather than a substitute for calibration.

score differences should be estimated using calendar-block or loan-aware resampling rather than treating 4.1 million correlated loan-months as independent. Third, the near-saturated HMM must be stress-tested against alternative estimation windows, covariance restrictions, state-orientation rules, and factor scaling. These requirements are part of the empirical result, not optional presentation refinements.

Bibliography

- Altman, E. I. (1968). Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. *The Journal of Finance*, 23(4):589–609.
- Bangia, A., Diebold, F. X., Kronimus, A., Schagen, C., and Schuermann, T. (2002). Ratings migration and the business cycle, with application to credit portfolio stress testing. *Journal of Banking & Finance*, 26(2–3):445–474.
- Basel Committee on Banking Supervision (2004). QIS 3: Definition of default. Basel II Quantitative Impact Study.
- Baum, L. E., Petrie, T., Soules, G., and Weiss, N. (1970). A maximization technique occurring in the statistical analysis of probabilistic functions of markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171.
- Bellotti, T. and Crook, J. (2009). Credit scoring with macroeconomic variables using survival analysis. *Journal of the Operational Research Society*, 60(12):1699–1707.
- Bellotti, T. and Crook, J. (2013). Retail credit stress testing using a discrete hazard model with macroeconomic factors. *Journal of the Operational Research Society*, 65(3):340–350.
- Bilmes, J. A. (1998). A gentle tutorial of the EM algorithm and its application to parameter estimation for Gaussian mixture and hidden markov models. Technical Report TR-97-021, International Computer Science Institute, Berkeley, CA.
- Board of Governors of the Federal Reserve System (2006). Basel ii: Definition of default.

- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3.
- Brookfield, H. A., Thornton, A. S., and Harrington, O. D. (2026). Macroeconomic-sensitive credit risk forecasting for consumer finance using regime-switching models. Preprints.org.
- Brown, I. and Mues, C. (2012). An experimental comparison of classification algorithms for imbalanced credit scoring data sets. *Expert Systems with Applications*, 39(3):3446–3453.
- Bussmann, N., Giudici, P., Marinelli, D., and Papenbrock, J. (2021). Explainable machine learning in credit risk management. *Computational Economics*, 57(1):203–216.
- Campbell, J. Y. and Cocco, J. F. (2015). A model of mortgage default. *The Journal of Finance*, 70(4):1495–1554.
- Costa e Silva, E., Lopes, I. C., Correia, A., and Faria, S. (2020). A logistic regression model for consumer default risk. *Journal of Applied Statistics*, 47(13–15):2879–2894.
- Crook, J. N., Edelman, D. B., and Thomas, L. C. (2007). Recent developments in consumer credit risk assessment. *European Journal of Operational Research*, 183(3):1447–1465.
- Davis, J. and Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 233–240.
- Deng, Y., Quigley, J. M., and Van Order, R. (2000). Mortgage terminations, heterogeneity and the exercise of mortgage options. *Econometrica*, 68(2):275–307.
- Dimitriadis, T., Gneiting, T., and Jordan, A. I. (2020). Evaluating probabilistic classifiers: Reliability diagrams and score decompositions revisited.
- Duffie, D., Saita, L., and Wang, K. (2007). Multi-period corporate default prediction with stochastic covariates. *Journal of Financial Economics*, 83(3):635–665.

- Duffie, D. and Singleton, K. J. (2003). *Credit Risk: Pricing, Measurement, and Management*. Princeton University Press.
- Elul, R., Souleles, N. S., Chomsisengphet, S., Glennon, D., and Hunt, R. M. (2010). What triggers mortgage default? *American Economic Review*, 100(2):490–494.
- Federal Reserve Bank of St. Louis (2026). FRED economic data. Monthly macroeconomic data series.
- Figlewski, S., Frydman, H., and Liang, W. (2012). Modeling the effect of macroeconomic factors on corporate default and credit rating transitions. *International Review of Economics & Finance*, 21(1):87–105.
- Fonseca, P. G. and Lopes, H. D. (2017). Calibration of machine learning classifiers for probability of default modelling. arXiv:1710.08901.
- Freddie Mac (2026). Single-family loan-level dataset. Public loan-level origination and monthly performance data.
- Fuster, A., Goldsmith-Pinkham, P., Ramadorai, T., and Walther, A. (2022). Predictably unequal? the effects of machine learning on credit markets. *The Journal of Finance*, 77(1):5–47.
- Gerardi, K., Herkenhoff, K. F., Ohanian, L. E., and Willen, P. S. (2018). Can’t pay or won’t pay? unemployment, negative equity, and strategic default. *The Review of Financial Studies*, 31(3):1098–1131.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.

- Hand, D. J. and Henley, W. E. (1997). Statistical classification methods in consumer credit scoring: A review. *Journal of the Royal Statistical Society: Series A*, 160(3):523–541.
- Jarrow, R. A. and Turnbull, S. M. (1995). Pricing derivatives on financial securities subject to credit risk. *The Journal of Finance*, 50(1):53–85.
- Khandani, A. E., Kim, A. J., and Lo, A. W. (2010). Consumer credit-risk models via machine-learning algorithms. *Journal of Banking & Finance*, 34(11):2767–2787.
- Kundig, P. and Sigrist, F. (2025). A spatio-temporal machine learning model for mortgage credit risk: Default probabilities and loan portfolios. *European Journal of Operational Research*.
- Lando, D. (1998). On Cox processes and credit risky securities. *Review of Derivatives Research*, 2(2–3):99–120.
- le Cessie, S. and van Houwelingen, J. C. (1992). Ridge estimators in logistic regression. *Applied Statistics*, 41(1):191–201.
- Lessmann, S., Baesens, B., Seow, H.-V., and Thomas, L. C. (2015). Benchmarking state-of-the-art classification algorithms for credit scoring: An update of research. *European Journal of Operational Research*, 247(1):124–136.
- Merton, R. C. (1974). On the pricing of corporate debt: The risk structure of interest rates. *The Journal of Finance*, 29(2):449–470.
- Milidonis, A. and Chisholm, K. (2024). The regime-switching structural default risk model. *Risks*, 12(3):48.
- Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600.
- Noriega, J. P., Rivera, L. A., and Herrera, J. A. (2023). Machine learning for credit risk prediction: A systematic literature review. *Data*, 8(11):169.

- Owusu-Manu, D.-G., Asiedu, R. O., Edwards, D. J., Donkor-Hyiaman, K., Abuntori, P. A., and El-Gohary, H. (2019). An assessment of mortgage loan default propensity in ghana. *Journal of Engineering, Design and Technology*.
- Rabiner, L. R. (1989). A tutorial on hidden markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence*, 1(5):206–215.
- Saito, T. and Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432.
- Shi, S., Tse, R., Luo, W., D’Addona, S., and Pau, G. (2022). Machine learning-driven credit risk: A systemic review. *Neural Computing and Applications*, 34(17):14327–14339.
- Shumway, T. (2001). Forecasting bankruptcy more accurately: A simple hazard model. *The Journal of Business*, 74(1):101–124.
- Simons, D. and Rolwes, F. (2009). Macroeconomic default modeling and stress testing. *International Journal of Central Banking*, 5(3).
- Tasche, D. (2013). The art of probability-of-default curve calibration. *Journal of Credit Risk*, 9(4), 63–103.
- Thomas, L. C. (2000). A survey of credit and behavioural scoring: Forecasting financial risk of lending to consumers. *International Journal of Forecasting*, 16(2):149–172.
- Tracy, J. and Wright, J. (2016). Payment changes and default risk: The impact of refinancing on expected credit losses. *Journal of Urban Economics*, 93:60–70.
- Van Calster, B., McLernon, D. J., van Smeden, M., Wynants, L., and Steyerberg, E. W. (2019). Calibration: The achilles heel of predictive analytics. *BMC Medicine*, 17(1):230.

- Van Calster, B., Nieboer, D., Vergouwe, Y., De Cock, B., Pencina, M. J., and Steyerberg, E. W. (2016). A calibration hierarchy for risk models was defined: From utopia to empirical data. *Journal of Clinical Epidemiology*, 74:167–176.
- Yu, F.-H., Lu, J., Gu, J.-W., and Ching, W.-K. (2018). Modeling credit risk with hidden markov default intensity. *Computational Economics*, 54(3):1213–1229.